

onlinePermOOB with LLM routing

Setup. `onlinePermOOB_with_LLM(df, ...)`. Last column of `df` is Y , never a feature. high-VIMP columns go to a frozen RF, low-VIMP columns go to a TabPFN-style k NN. Fit once on the reference window. Trail batches score $T_t = \text{MSE}_t - E_{\text{ref}}$. ADDIS is the primary mark.

Table 1: False-alarm rate, no labeled onset. high-VIMP RF + low-VIMP TabPFN-style k NN. Share of 12 replications with at least one rejection over 40 batches. ADDIS is the primary OnlineRFPerm mark.

method	stationary FAR (%)	random-noise FAR (%)
OnlineRFPerm + ADDIS	0.0	0.0
OnlineRFPerm + SAFFRON	0.0	0.0
OnlineRFPerm + fix- α	41.7	33.3
last-two hop ($1.5\times$)	100.0	100.0
Page-Hinkley	58.3	66.7
EWMA	0.0	8.3
CUSUM	75.0	41.7
DDM	8.3	16.7
ADWIN-lite	0.0	25.0
betting martingale	0.0	0.0

Table 2: `onlinePermOOB_with_LLM` on the tables we have. Last column is Y . high-VIMP RF + low-VIMP TabPFN-style k NN. Entries are first rejection batch ($-1 = \text{none}$).

table	n	p	ADDIS	SAFFRON	PH	ADWIN
eeg	4000	14	-1	-1	17	17
electricity	4000	7	-1	-1	13	-1
bankmarketing	4000	16	-1	-1	17	-1
covertype	4000	54	-1	-1	12	18